

CODE SPORT



In this month's column, we discuss a few interview questions related to machine learning and data science.

As we have been doing over the last couple of months, we will continue to discuss a few more computer science interview questions in this column as well, particularly focusing on topics related to data science, machine learning and natural language processing. It is important to note that many of the questions are typically oriented towards practical implementation or deployment issues, rather than just concepts or theory. So it is important for interview candidates to make sure that they get adequate implementation experience with machine learning/ NLP projects before their interviews. Data science platforms such as Kaggle (www.kaggle.com) host a number of competitions that candidates can attempt to practice their skills on. Also, many of the data science or machine learning related academic computer conferences host data challenge competitions such as the KDD Cup (<http://www.kdd.org/kdd-cup>). Data science enthusiasts can sign on for these challenges and hone their skills in solving real life problems. Let us now discuss a few interview questions.

1. You are given 100,000 movie reviews that are labelled as positive or negative. You have been told to perform sentiment analysis on the new incoming reviews by classifying each review as positive or negative, which is a simple binary classification problem. Can you explain what features you would use for this classification problem? Once you decide on your set of features, how would you go about selecting which classifier to use?
2. Let us assume that you decided to use the 'bag of words' approach in the above problem with each vocabulary term becoming a feature for your classifier. Essentially, you can construct a feature set where the dimensions of this set are the same as the size of your vocabulary, and each feature corresponds to a specific term in the vocabulary. The feature value can either be the count of the term or merely the presence or absence of the

term in the document; or you can employ Tf-Idf count for each term-review combination, etc.

You had used a random forests classifier for sentiment classification. Now you are told that your vocabulary size is 100,000. Would this change your decision about which classifier to use?

3. For problem (1), you had decided to use a support vector machine classifier. However, now you are told that instead of just doing binary classification of the reviews, you need to classify them as one of five categories, namely: (a) strongly positive, (b) weakly positive, (c) neutral, (d) weakly negative, and (e) strongly negative. You are given labelled data with these five categories now. Would you still continue to use the 'support vector machine' (SVM) classifier? If so, can you explain how SVM handles multi-class classification? If you decide to switch from SVM to a different classifier, explain the rationale behind your switch.
4. For the sentiment classification problem, other than the review text itself, you are now given additional data about the movies. This additional data includes the reviewers' names, address, age, country of residence, date of review and the specific movie genre they are interested in. This additional data contains both numeric and string data, with some of the features being categorical. A country's name is string data, and the movie genre is string data which is actually categorical. What kind of data preprocessing would you do on this additional data to use it with your classifier?
5. Generally, interviewers expect you to be familiar with some of the popular libraries that can be used for data science. So some of the questions can be library-specific as well. In question (4), you may be asked to mention how you would convert categorical data to numeric form. Can you write a piece of Python code to do this conversion?
6. Let us assume that you decided to use a SVM